

HOMEWORK #3

ECBM E4070/6070, Professor Aurel A. Lazar

Deadline: 11:59PM (Midnight), Friday, March 6, 2026

You are given a notebook, `homework3_handout.ipynb`, to aid your code organization; please follow the Homework policies addressed in the handout as well. Please post any questions regarding this assignment on the **Ed Discussion** forum. Your questions and the answers may also benefit the other students in class.

Task 1: End-to-end Motion Detection Circuit (10pt)

You will explore motion detection models, specifically the *Hassenstein-Reichardt Motion Detector* (HR Motion Detector, Figure 1) and the *Barlow-Levick Inspired Motion Detector* (BL-Inspired Motion Detector, Figure 2). We will be using the *NeuroCircuitDesk* (NCD) package to simulate end-to-end the *Drosophila* early vision system, from the sampling in the retina to motion detection in the medulla following the intrinsic geometry known as *retinotopy*.

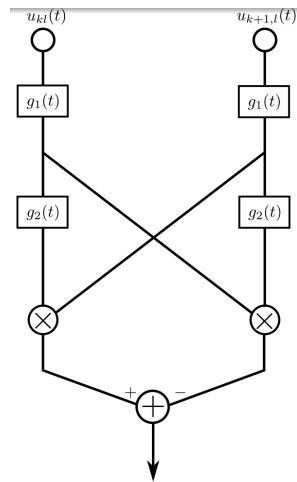


Figure 1: The Hassenstein-Reichardt motion detector

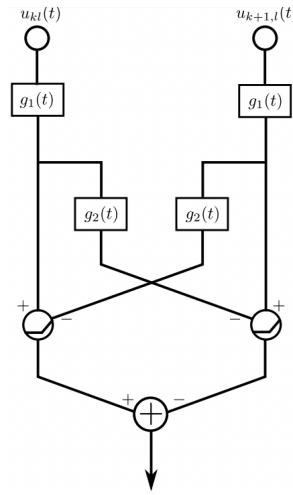


Figure 2: The Barlow-Levick Inspired motion detector

Task 1.1: Install NeuroCircuitDesk

Please refer to `homework3_supplement.pdf` in the attached `homework3_supplements.zip` file for the NCD installation process.

Task 1.2: Implement the Motion Detector

Section 1.1 of the notebook defines the MicroCircuits (MCs) for each of the stages in the early vision system circuit. The *motion_detector_template*, a multi-input-multi-output (MIMO) MC, models the T4 and T5 neurons. An instance of this MC is centered at each retinotopic column, receiving inputs from *seven* columns that are locally ordered as illustrated in Figure 3. To account for the four subtypes (a, b, c, d) of each motion detection neuron, each MC will have four outputs corresponding to the motion response in the four cardinal directions.

Select and implement one of the HR or BL detectors, shown in Figure 1 and 2 respectively:

- Complete the *motion_detection_algorithm* function in Section 1.2 of the notebook highlighted with "TODO" comment. Use $g_1(t) = \delta(t)$ (i.e., no high pass filter) and $g_2(t) = \delta(t - \tau)$ (τ is a delay parameter, which is the only parsed parameter of the algorithm).
- As a single HR detector has only one preferred direction (left to right in Figure 1), you will implement 4 instances of HR detector within the algorithm using 4 different sets of inputs, and assign them to the four outputs.

The MCs are connected to compose the complete end-to-end circuit in Section 1.3. A natural scene video is provided as an *.h5* file, which we will use as input to the circuit. Compile and execute the circuit, and visualize the output for your motion detector using the provided code.

(5pt) For bonus points, implement the other detector algorithm. (You may need to rerun notebook section 1.3 to build a new circuit).

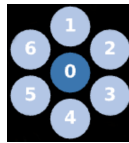


Figure 3: Local order of input columns to a *motion_detector_template* MC.

Task 2: Ring-attractor Model of Ellipsoid Body

In this question, you will implement a simplified model for the bump rotation circuit, depicted in Figure 4. The 16 EPG neurons are assigned to 8 EB wedges, with two neurons in each wedge. Each PEN neuron is assigned to one of 16 PB glomeruli, with 8 glomeruli in the left and right halves of PB. Each EB wedge receives input from two PEN neurons, one each in the left and right halves of PB. To rotate the EPG bump anticlockwise or clockwise respectively, the 8 PEN neurons in the left and right half of PB are

respectively excited, while the other 8 neurons receive no input. The set of Delta7 local neurons in the PB have been replaced by a single Delta7 neuron for simplicity. This Delta7 neuron receives equal input from, and equally inhibits all 16 EPG neurons.

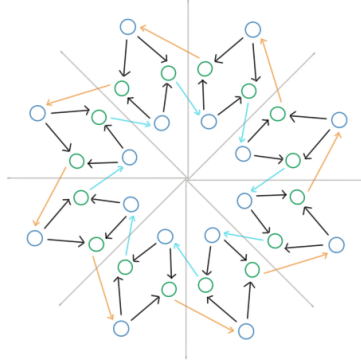


Figure 4: The simplified bump rotation circuit. The EPG and PEN neurons are respectively denoted by green and blue nodes. Note that Delta7 neurons have been omitted. The yellow and light-blue synaptic connections respectively enable counterclockwise and clockwise rotation of the EPG bump.

Each neuron is modeled as a discrete dynamical system, and the “firing rate” $x[n]$ at time-step n for synaptic inputs $\{I_i[n]\}$ is given by the equation $x[n] = h(\alpha x[n-1] + \sum_i \beta_i I_i[n])$. Here $h(\cdot)$ is the Clipped Rectified Linear Unit which clips the output to lie between 0 and 1. Our simplified discrete-time neuron model retains many essential functional aspects of a typical continuous-time neuron: $\alpha x[n-1]$ resembles the leak term, $\{\beta_i\}$ are the set of synaptic weights for inputs $\{I_i\}$. Clipping the output below 0 emulates a neuron’s firing threshold, while clipping the output above 1 emulates the saturating effect of a neuron’s refractory period on its firing rate.

Tasks

1. (4 points) Use NeuroNLP on the Hemibrain dataset to explore the actual morphology of the circuit. Attach screenshots of the following: (i) All the EL neurons (ii) Pick on EL neuron and add its EPG partners. Do you indeed observe that one EL neuron is connected to two EPG neurons on average? (iii) Pick two neighboring EPG neurons and add partner PEN neurons. Do you indeed observe that on average one PEN neuron connects to two EPG neurons?
2. (6 points) Implement the bump rotation circuit described above and iteratively solve the discrete-time dynamical equations for all neurons in the circuit. You can use $\alpha = 0.15$ for all neurons, and need to find appropriate values for all synaptic weights $\{\beta\}$. Visualize activations of the 16 EPG and PEN neurons over time and demonstrate the EPG bump rotating (at least) 360 degrees in response to (i) clockwise (ii) counterclockwise angular velocity input.

Note: You are allowed to add additional constant input bias currents to neurons, wherein one of the input currents I_i is any constant real number. It is instructive to think about how adding such a bias current “shifts” the output response curve of a neuron, and why this may be needed for some neurons to function as desired.